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### Fabrication of printed fuse

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#### Abstract

A power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system is provided. The power fuse includes at least one fuse element assembly that includes one or more substrates, one or more sets of weak spots, and a conductor. The weak spots are formed on the substrates, and the substrates are longitudinally spaced apart from one another along the conductor. The conductor is separately provided from the substrate and the weak spots. The conductor includes one or more strips of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to the transient load current cycling events. The conductor includes connector sections that are attached to respective ones of the sets of weak spots, and extending sections coupling the connector sections.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 17/185,537 filed on Feb. 25, 2021, entitled “Design and Fabrication of Printed Fuse,” which is a continuation-in-part under 35 U.S.C. § 120 of U.S. patent application Ser. No. 16/590,020 filed on Oct. 1, 2019, entitled “Design and Fabrication of Printed Fuse,” now U.S. Pat. No. 11,087,943, issued on Aug. 10, 2021, which relates in subject matter to and claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Ser. No. 62/897,024 filed Sep. 6, 2019, entitled “Design and Fabrication of Printed Fuse,” the entirety of all which documents are incorporated by reference.

### BACKGROUND

(1) The field of the disclosure relates generally to electrical circuit protection fuses, and more specifically to the fabrication of power fuses including thermal-mechanical strain fatigue resistant fusible element assemblies.

(2) Fuses are widely used as overcurrent protection devices to prevent costly damage to electrical circuits. Fuse terminals typically form an electrical connection between an electrical power source or power supply and an electrical component or a combination of components arranged in an electrical circuit. One or more fusible links or elements, or a fuse element assembly, is connected between the fuse terminals, so that when electrical current flowing through the fuse exceeds a predetermined limit, the fusible elements melt and open one or more circuits through the fuse to prevent electrical component damage.

(3) Full-range power fuses are operable in high voltage power distributions to safely interrupt both relatively high fault currents and relatively low fault currents with equal effectiveness. In view of constantly expanding variations of electrical power systems, known fuses of this type are disadvantaged in some aspects. Improvements in full-range power fuses are desired to meet the needs of the marketplace.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting and non-exhaustive embodiments are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various drawings unless otherwise specified.

(2) FIG. 1 illustrates an exemplary transient current pulse profile generated in an electrical power system.

(3) FIG. 2A illustrates a schematic perspective view of a known power fuse.

(4) FIG. 2B is a schematic perspective view of the fuse element assembly of the power fuse illustrated in FIG. 2A.

(5) FIG. 2C is a schematic diagram of a weak spot of the fuse element assembly illustrated in FIG. 2B.

(6) FIG. 2D is a schematic diagram illustrating the weak spots of the fuse element assembly illustrated in FIG. 2B under load current cycling events.

(7) FIG. 2E is a schematic diagram illustrating the weak spots of the fuse element assembly illustrated in FIG. 2E fail after load current cycling events.

(8) FIG. 3 is a schematic partial perspective view of an exemplary power fuse.

(9) FIG. 4 is a schematic enlarged view of the fuse element assembly for the power fuse illustrated in FIG. 3.

(10) FIG. 5 illustrates the substrate and weak spots of the fuse element assembly illustrated in FIG.

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(11) FIG. 6 is a schematic cross-sectional magnified view of a portion of an exemplary fuse element assembly.

(12) FIG. 7 is a schematic diagram illustrating the arcing in the fuse element assembly illustrated in FIG. 4.

(13) FIG. 8 is a schematic diagram of an exemplary method for fabricating the power fuse illustrated in FIGS. 3-7.

(14) FIG. 9 is a schematic flow chart illustrating the method illustrated in FIG. 8.

(15) FIG. 10A is a schematic perspective view of another exemplary fuse element assembly for the power fuse illustrated in FIG. 3.

(16) FIG. 10B is a schematic cross-sectional view of the fuse element assembly illustrated in FIG. 10A along line 10B-10B.

(17) FIG. 10C is a schematic enlarged cross-sectional view of a portion of the fuse element assembly illustrated in FIG. 10A.

(18) FIG. 11 is a schematic flow chart of another exemplary method of fabricating a power fuse.

(19) FIG. 12 is a schematic diagram of attaching the conductor with the weak spots in the method illustrated in FIG. 11.

(20) FIG. 13A is a schematic top view of one more exemplary fuse element assembly for the power fuse illustrated in FIG. 3.

(21) FIG. 13B is a schematic cross-sectional view of the fuse element assembly illustrated in FIG. 13A along line 13B-13B.

(22) FIG. 13C is a schematic cross-sectional view of the fuse element assembly illustrated in FIG. 13A along line 13B-13B, where the fuse element has an alternative configuration from that of the fuse element illustrated in FIG. 13B.

(23) FIG. 14 is a schematic diagram of an exemplary coupling mechanism between the weak spot and the conductor of the power fuse illustrated in FIG. 3.

#### DETAILED DESCRIPTION

(24) Recent advancements in electric vehicle technologies present unique challenges to fuse manufacturers. Electric vehicle manufacturers are seeking fusible circuit protection for electrical power distribution systems operating at voltages much higher than conventional electrical power distribution systems for vehicles, while simultaneously seeking smaller fuses to meet electric vehicle specifications and demands

(25) Electrical power systems for conventional, internal combustion engine-powered vehicles operate at relatively low voltages, typically at or below about 48 VDC. Electrical power systems for electric-powered vehicles, referred to herein as electric vehicles (EVs), however, may operate at much higher voltages. The relatively high voltage systems (e.g., 200 VDC and above) of EVs may generally enable the batteries to store more energy from a power source and provide more energy to an electric motor of the vehicle with lower losses (e.g., heat loss) than conventional batteries storing energy at 12 Volts (V) or 24 V used with internal combustion engines, and more recent 48 V power systems.

(26) EV original equipment manufacturers (OEMs) employ circuit protection fuses to protect electrical loads in all-battery electric vehicles (BEVs), hybrid electric vehicles (HEVs) and plug-in hybrid electric vehicles (PHEVs). Across each EV type, EV manufacturers may seek to maximize the mileage range of the EV per battery charge while reducing cost of ownership. Accomplishing these objectives turns on the energy storage and power delivery of the EV system, as well as the size, volume, and mass of the vehicle components that are carried by the power system. Smaller and/or lighter vehicles may more effectively meet these demands than larger and heavier vehicles. As such, all EV components are now being scrutinized for potential size, weight, and cost savings.

(27) Generally speaking, larger components may tend to have higher associated material costs, tend to increase the overall size of the EV or occupy an undue amount of space in a shrinking vehicle

volume, and may tend to introduce greater mass that directly reduces the vehicle mileage per single battery charge. Known high voltage circuit protection fuses may, however, be relatively large and relatively heavy components. Historically, and for good reason, circuit protection fuses have tended to increase in size to meet the demands of high voltage power systems as opposed to lower voltage systems. As such, existing fuses needed to protect high voltage EV power systems may be much larger than the existing fuses needed to protect the lower voltage power systems of conventional, internal combustion engine-powered vehicles. Smaller and lighter high voltage power fuses are desired to meet the needs of EV manufacturers, without sacrificing circuit protection performance. (28) Electrical power systems for state of the art EVs may operate at voltages as high as 450 VDC or even higher. The increased power system voltage may desirably deliver more power to the EV per battery charge. Operating conditions of electrical fuses in such high voltage power systems may be much more severe, however, than lower voltage systems. Specifically, specifications relating to electrical arcing conditions when the fuse opens may be particularly difficult to meet for higher voltage power systems, especially when coupled with the industry preference for reduction in the size of electrical fuses. Current cycling loads imposed on power fuses by state of the art EVs may also tend to impose mechanical strain and wear that can lead to premature failure of a conventional fuse element. While known power fuses may be presently available for use by EV OEMs in high voltage circuitry of state of the art EV applications, the size and weight, not to mention the cost, of conventional power fuses capable of meeting the requirements of high voltage power systems for EVs can be impractically high for implementation in new EVs.

(29) Providing relatively smaller power fuses that can capably handle high current and high battery voltages of state of the art EV power systems, while still providing acceptable interruption performance as the fuse element operates at high voltages can be challenging, to say the least. Improvements are needed to longstanding and unfulfilled needs in the art.

(30) While described in the context of EV applications and a particular type and ratings of fuse, the benefits of the disclosure are not necessarily limited to EV applications or to the particular type or ratings described. Rather, the benefits of the disclosure are believed to more broadly accrue to many different power system applications and can also be practiced in part or in whole to construct different types of fuses having similar or different ratings than those discussed herein.

(31) FIG. 1 illustrates an exemplary current drive profile **100** in an EV power system application that can render a fuse, and specifically the fuse element or elements therein susceptible to load current cycling fatigue. The current is illustrated along a vertical axis in FIG. 1 with time illustrated along the horizontal axis. In typical EV power system applications, power fuses may be used as circuit protection devices to prevent damage to electrical loads from electrical fault conditions. The power system may be operated at voltages above 500 V and/or at currents above 150 Amperes (A). Considering the example of FIG. 1, EV power systems experience large seemingly random variance in current loads over relatively short periods of time, for example, between -250 A and 150 A. The seemingly random variance in current produces current pulses of various magnitudes in sequences caused by seemingly random driving habits based on the actions of the driver of the EV vehicle, traffic conditions and/or road conditions. This creates a practically infinite variety of current loading cycles on the EV drive motor, the primary drive battery, and any protective power fuse included in the system.

(32) Such random current loading conditions, exemplified in the current pulse profile of FIG. 1, may be cyclic in nature for both the acceleration of the EV (corresponding to battery drain) and the deceleration of the EV (corresponding to regenerative battery charging). This current cyclic loading imposes thermal cycling stress on the fuse element, and more specifically in the weak spots of the fuse element assembly in the power fuse, by way of a joule effect heating process. This thermal cyclic loading of the fuse element imposes mechanical expansion and contraction cycles on the fuse element weak spots in particular. This repeated mechanical cyclic loading of the fuse element weak spots imposes an accumulating strain that damages the weak spots to the point of breakage

over time. For the purposes of the present description, this thermal-mechanical process and phenomena is referred to herein as fuse fatigue. As explained further below, fuse fatigue is attributable mainly to creep strain as the fuse endures the drive profile. Heat generated in the fuse element weak spots is the primary mechanism leading to the onset of fuse fatigue.

(33) FIG. 2A shows a known high voltage power fuse **200** that is designed for use with an EV power system. The power fuse **200** includes a housing **202**, terminal blades **204**, **206** configured to connect to line and load side circuitries, and a fuse element assembly **208** that completes an electrical connection between the terminal blades **204**, **206** through terminal contact blocks **222**, **224** provided on end plates **226**, **228**. When subjected to predetermined current conditions, at least a portion of the fuse element assembly **208** melts, disintegrates, or otherwise structurally fails and opens the circuit path between the terminal blades **204**, **206**. Load side circuitry is therefore electrically isolated from the line side circuitry to protect load side circuit components from damage when electrical fault conditions occur.

(34) FIG. 2B illustrates the fuse element assembly **208** in further detail. The fuse element assembly **208** is generally formed from a strip of electrically conductive material into a series of co-planar sections **240** connected by oblique sections **242**, **244**. The oblique sections **242**, **244** may be formed or bent out of plane from the planar sections **240**.

(35) In the example illustrated, the planar sections **240** define a plurality of sections of reduced cross-sectional area **241**, referred to in the art as weak spots. The weak spots **241** may be defined by apertures in the planar sections **240**. The weak spots **241** correspond to the narrow portion of the section **240** between adjacent apertures. The reduced cross-sectional areas at the weak spots **241** will experience higher heat concentration than the rest of the fuse element assembly **208** as current flows through the fuse element assembly **208**.

(36) The weak spots **241** of the fuse element assembly **208** fabricated by metal stamping or punching have been found to be disadvantageous for EV applications having the type of cyclic current loads described above. Such stamped fuse element designs undesirably introduce mechanical strains and stresses on the fuse element weak spots **241** such that a shorter service life tends to result. This short fuse service life manifests itself in the form of nuisance fuse operation resulting from the mechanical fatigue of the fuse element at the weak spots **241**.

(37) FIG. 2C shows the cross-sectional view of a metal plate **250** after an aperture **252** is punched through the metal plate **250**. After a punching or stamping process, micro tears **254** occur along the border **256** of the aperture **252**.

(38) As illustrated in FIGS. 2D and 2E, the weak spots **241** of the fuse element assembly **208** experience repeated high current pulses and cyclic current events (FIG. 2D), which lead to metal fatigue from grain boundary disruptions followed by crack propagation and failure in the fuse element assembly **208** at the weak spots **241** (FIG. 2E). The mechanical constraints of the fuse element assembly **208** may be inherent in the stamped fuse element design and manufacture, which unfortunately has been found to promote in-plane buckling of the weak spots **241** during repeated load current cycling. This in-plane buckling is the result of damage to the metal grain boundaries where a separation or slippage occurs between adjacent metal grains. Such buckling of weak spots **241** occurs over time and is accelerated and more pronounced with higher transient current pulses. The greater the heating-cooling delta in the transient current pulses the greater the mechanical influence, and thus the greater the in-place buckling deformation of the weak spots **241**.

(39) Repeated physical mechanical manipulations of metal, caused by the heating effects of the transient current pulses, in turn cause changes in the grain structure of metal fuse element. These mechanical manipulations may be sometimes referred to as working the metal. Working of metals will cause a strengthening of the grain boundaries where adjacent grains may be tightly constrained to neighboring grains. Over working of a metal will result in disruptions in the grain boundary, where grains slip past each other and cause what is called a slip band or plane. This slippage and separation between the grains result in a localized increase of the electrical resistance that

accelerates the fatigue process by increasing the heating effect of the current pulses. The formation of slip bands is where fatigue cracks are first initiated.

(40) The inventors have found that a manufacturing method of stamping or punching metal to form the fuse element assembly **208** causes localized slip bands on all stamped edges of the fuse element weak spots **241** because the stamping processes to form the weak spots **241** may be shearing and tearing mechanical processes. This tearing process pre-stresses the weak spots **241** with many slip band regions. The slip bands and fatigue cracks, combined with the buckling described due to heat effects, eventually lead to a premature structural failure of the weak spots **241** that may be unrelated to electrical fault conditions. Such premature failure mode that does not relate to a problematic electrical condition in the power system is sometimes referred to as nuisance operation of the fuse. Since once the fuse elements fail the circuitry connected to the fuse is not operational again until the fuse is replaced, avoiding such nuisance operation is highly desirable in an EV power system from the perspective of both EV manufacturers and consumers. Indeed, given an increased interest in EV vehicles and their power systems, the effects of fuse fatigue may be deemed to be a negative Critical to Quality (CTQ) attribute in the vehicle design.

(41) Accordingly, improved fuse elements and methods for fabricating fuse elements including weak spots that are fatigue resistant may be highly desirable.

(42) Exemplary embodiments of fuse elements and the method of fabricating such fuse elements are described below that advantageously avoid the strain damages at weak spots from the manufacturing process of stamping or punching, while also providing an effective arc extinguishing mechanism. Weak spots in the exemplary embodiments may be formed directly onto a planar substrate, avoiding micro tears from the punching or stamping processes. The weak spots may be connected by a separately-fabricated conductor having coplanar connector sections and oblique connector sections used for effective arc extinguishing.

(43) While described below in reference to particular embodiments, such description is intended for the sake of illustration rather than limitation. The significant benefit of the inventive concepts will now be explained in reference to the exemplary embodiments illustrated in the FIGS. Method aspects will be in part apparent and in part explicit in the following discussion.

(44) Referring now to FIGS. **3-7**, an exemplary power fuse **300** is illustrated. The power fuse **300** includes at least one fuse element assembly **302** (FIG. **3**). The power fuse **300** may include a housing **308**. The power fuse **300** further includes terminal blades **304**, **306** configured to connect the power fuse **300** to line and load side circuitry. The electrical connection of the fuse element assembly **302** is completed through terminal contact blocks **322**, **324** provided on end plates **332**, **334** and the terminal blades **304**, **306**. When subjected to predetermined current conditions, at least a portion of the fuse element assembly **302** melts, disintegrates, or otherwise structurally fails and opens the circuit path between the terminal blades **304**, **306**. Load side circuitry is therefore electrically isolated from the line side circuitry to protect load side circuit components from damage when electrical fault conditions occur.

(45) FIG. **4** shows the exemplary fuse element assembly **302** in further detail. The fuse element assembly **302** includes a substrate **310**, a plurality of weak spots **312**, and a conductor **314**.

(46) The substrate **310** may be a planar substrate (FIG. **5**). The substrate **310** may be elongated. In the exemplary embodiment, the top surface of the substrate **310** is rectangular. In some embodiments, the substrate **310** is ceramic. In one example, the substrate is alumina ceramic. An alumina substrate has a relatively high thermal conductivity (e.g., approximately 30 Wm.sup.-1K.sup.-1), which helps dissipate heat from the weak spots **312**.

(47) In the exemplary embodiment, the weak spots **312** may be formed on the substrate **310**. The number of weak spots **312** can be three or other numbers such as one, two, or four that enable the fuse element assembly **302** to function as described herein. The weak spots **312** may be spaced apart from each other. In some embodiments, the weak spots **312** may be disposed apart from each other along the longitudinal direction of the substrate **310**. The weak spots **312** may be made of

conductive material such as copper. The weak spots **312** may be printed on the substrate **310** using known techniques. In some embodiments, however, the weak spots **312** may be formed on the substrate **310** using techniques other than printing. Multiple layers of the weak spots **312** may be formed over one another to change the overall thickness of the weak spots **312**. The electrical resistance and performance of the weak spots **312** may be, therefore, relatively more controllable than the weak spots formed by metal stamping or punching. Because the weak spots **312** may be formed without mechanical micro tears from the mechanical manufacturing processes like metal stamping or punching, the weak spots **312** do not suffer from load current cycling fatigue as the weak spots **241** of the known fuse **200**, especially when under the large, seemingly random cyclic current changes in a direct current power system of an EV.

(48) In some embodiments, the fuse element assembly **302** further includes a dielectric layer **316** disposed between the substrate **310** and the weak spots **312** (FIG. 6). In an exemplary embodiment, the dielectric layer **316** may be glass or another suitable dielectric material known in the art. When weak spots **312** may be formed with only electrically-conductive materials, the electrically-conductive materials separate when the materials melt in a fusing condition but may reconnect thus allowing the circuit to reconnect. To minimize this reconnection of weak spots **312** to allow the power fuse **300** to function at predetermined current conditions, a layer of dielectric, glass-based layer **316** is deposited under the weak spot **312**. The material for the dielectric layer **316** is selected such that it melts at a higher temperature than the weak spots **312** but at a low enough temperature that allows diffusion. The melting temperature of the dielectric layer **316** is approximately 25° C.-50° C. above the maximum fusing temperature of the weak spots **312**. This temperature range allows the dielectric layer **316** to be mechanically stable during the fusing process to support the weak spots **312** while allowing the dielectric material to diffuse into the weak spots **312**. The melting temperature of the dielectric layer **316** may vary depending on materials. The diffusion is desired for two reasons. First, it provides a means to adjust the weak spot resistance, where more fusing results in more diffusion and higher resistivity. Second, the diffused dielectric layer **316** changes the wetting characteristics of the conductor and does not allow the melted weak spots **312** to reattach.

(49) Referring back to FIG. 4, the weak spots **312** of the fuse element assembly may be connected through the conductor **314**. In the exemplary embodiments, the conductor **314** is made from a solid elongated strip metal. The conductor **314** may be made by punching or stamping a solid elongated strip metal. The thickness of the conductor **314** is greater than the weak spots **312**. As a result, the weak spots **312** experience more heat than the conductor **314** and open before the conductor **314** under predetermined current conditions. The conductor **314**, therefore, does not have stamped weak spot openings and avoids thermal-mechanical fatigue strain when subjected to transient load current cycling events.

(50) In an exemplary embodiment, the conductor **314** includes coplanar connector sections **318** and obliquely extending sections **320**. The obliquely extending sections **320** bend out of plane of the coplanar connector sections **318**. The conductor **314** may further include first and second terminal tabs extending from the obliquely extending sections **320**. The conductor **314** couples to terminal contact blocks **322**, **324** through the terminal tabs **326**, **328**.

(51) In the contemplated embodiment, the coplanar connector sections **318** may be mounted on respective ones of the weak spots **312**. Alternatively, the coplanar connector sections **318** may be mounted on the substrate **310** and may be connected with weak spots **312**. As a result, the obliquely extending sections **320** extend above the substrate **310** in between the weak spots **312**, and the first and second terminal tabs **326**, **328** may extend coplanar to one another in a plane spaced from the coplanar connector sections **318** and the substrate **310**. The plane of the first and second terminal tabs **326**, **328** may extend parallel to the coplanar connector sections **318** and the substrate **310**.

(52) In the exemplary embodiment, the power fuse **300** includes three fuse element assemblies **302** (FIG. 3). The power fuse **300** may in other embodiments include other numbers of fuse element

assemblies **302**, such as one and two, that enable the power fuse **300** to function as described herein. The plurality of fuse element assemblies **302** may be connected in parallel with each other to increase the ratings of the power fuse **300** without increasing the physical size of the power fuse **300**. The fuse element assemblies **302** may be arranged such that two neighboring fuse element assemblies may be mirror images of each other. The fuse element assemblies **302** may be stacked together with the substrate of one fuse element assembly facing the conductor of another fuse element assembly.

(53) A full-range fuse can be realized by using at least one fuse element assembly **302** that is responsive to relatively low current operation (or overload faults) and at least one fuse element assembly **302** that is responsive to relatively high current operation (or short circuit faults). The fuse element assemblies **302** may also be used in a fuse that is not full range.

(54) In the exemplary embodiment, the power fuse **300** may further include an arc extinguishing filler **330** (FIG. 7). The arc extinguishing filler **330** surrounds at least part of the fuse element assembly **302**. The arc extinguishing filler **330** may be disposed underneath the obliquely extending sections **320**. The arc extinguishing filler **330** may also be disposed above the obliquely extending sections **320**, the coplanar connector sections **318**, and the weak spots **312**. The arc extinguishing filler **330** may be introduced to the housing **308** via one or more fill openings in one of the end plates **332**, **334** that may be sealed with plugs (not illustrated). The plugs may be fabricated from steel, plastic or other materials in various embodiments. In other embodiments a fill hole or fill holes may be provided in other locations, including but not limited to the housing **308** to facilitate the introduction of the arc extinguishing filler **330**.

(55) In one contemplated embodiment, the arc extinguishing filler **330** is composed of quartz silica sand and a sodium silicate binder. The quartz sand has a relatively high heat conduction and absorption capacity in its loose compacted state, but can be silicated to provide improved performance. For example, a liquid sodium silicate solution is added to the sand and then the free water is dried off. Separately provided arc barrier materials (not illustrated) may also be provided to prevent arcing from reaching the ends of the terminal tabs **326**, **328**.

(56) In the exemplary embodiment, the fuse element assembly **302** provides access of the arc to the arc quenching medium such as sand in the arc extinguishing filler **330**. When weak spots **312** melt at predetermined current conditions, arcing starts at weak spots **312**. As the arc grows in length it migrates from the weak spots **312** and the substrate **310** and follows the obliquely extending sections **320** into the surrounding arc extinguishing filler **330** for efficient cooling and quicker extinguishment.

(57) FIGS. **8** and **9** show an exemplary method **900** of fabricating a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system. FIG. **8** shows a schematic diagram of the method **900**, while FIG. **9** shows a flow chart of the method **900**. The method **900** includes forming **902** a plurality of fusible weak spots on a planar substrate such that the plurality of fusible weak spots may be longitudinally spaced from one another on the planar substrate. The method **900** further includes providing **904** a conductor separately from the planar substrate and the plurality of weak spots. The number of coplanar connector sections of the conductor may be the same as the number of weak spots formed on the planar substrate. The method **900** also includes **906** mounting the coplanar connector sections of the conductor to respective ones of the plurality of weak spots. As a result, the obliquely extending sections of the conductor extend above the elongated planar substrate in between the plurality of fusible weak spots, and the first and second terminal tabs of the conductor extend coplanar to one another in a plane parallel to but spaced from the coplanar connector sections and the substrate. In one example, the coplanar connection sections of the conductor may be brazed to the weak spots. In some embodiments, the conductor is formed in one piece. The conductor **800** may include support bridges **802** connecting the coplanar connector sections **318** (FIG. **8**). The method **900** may further include removing the support bridges after the coplanar connector sections of the conductor

have been mounted on respective ones of the plurality of weak spots.

(58) FIGS. **10A-10C** show another exemplary fuse element assembly **1302** included in the power fuse **300**. FIG. **10A** is a perspective view of the fuse element assembly **1302**. FIG. **10B** is a cross-sectional view of the fuse element assembly **1302** along line **10B-10B** illustrated in FIG. **10A**. FIG. **10C** is an enlarged view of a section of the fuse element assembly **1302** as marked in FIG. **10B**. Different from the fuse element assembly **302**, where the substrate **310** is in one piece (see FIG. **4**), substrates **1310** of the fuse element assembly **1302** may be separate from one another. The weak spot **1312** and its substrate **1310** may be collectively referred to as a weak spot pad **1203**. The substrate **310** typically is not electrically conductive, but may become electrically conductive in the DC arc plasma. Therefore, the arc may not be extinguished within a desired period of time, causing damages to the fuse element assembly **302** and the power fuse **300**.

(59) In contrast, the substrates **1310** of the fuse element assembly **1302** may be separate from one another. The arc extinguishing filler **330** may be filled in the fuse **300**, including the space separating adjacent substrates **1310**. As a result, the arc is suppressed by the arc extinguishing filler **330**.

(60) In the exemplary embodiment, weak spots **1312** may be included in the fuse element assembly **1302**. Compared to the weak spot **312** (see FIG. **5**), which is one block, the weak spot **1312** has one or more openings **1208** (FIG. **10A**, also see FIG. **12**). The openings **1208** increase the number of weak spots **1312** in one weak spot pad **1203**. Instead of one single weak spot **312**, the weak spot **1312** includes additional weak spots defined by the openings **1208**, besides the weak spot **1312** itself. The openings **1208** may be illustrated as rectangular in the depicted embodiment for the illustration purpose only. The openings **1208** may be in other shapes such as circular, oval, square, or irregular shapes that enable the openings **1208** to function as described herein. In some embodiments, the fuse element assembly **1302** includes the weak spots **312**, instead of the weak spots **1312**. In other embodiments, the fuse element assembly **1302** includes a mix of the weak spots **312** and the weak spots **1312**.

(61) In some embodiments, the substrate **1310** may be a rod having an increased thickness than a substrate formed as a sheet. The rod may be a square rod or rectangular rod where the axial profile **1020** is square or rectangular (FIG. **10B**). The axial profile **1020** may be in other shapes that enable the substrate **1310** to function as described herein. Compared to a sheet substrate, a rod substrate **1310** has an increased mass, which increases the thermal conductivity of the substrate **1310** for transmitting heat away from the conductor **314** and the weak spots **312**, **1312**.

(62) FIG. **11** is a flow chart of another exemplary method **1100** of fabricating a power fuse for protecting an electrical load subject to transient load current events in a direct current electrical power system. The method **1100** includes forming **1102** a plurality of fusible weak spots on a plurality of substrates. The plurality of fusible weak spots **1312** may be first formed on the substrate **1310** that is initially one single piece and the plurality of fusible weak spots **1312** may be then separated from one another by cutting the substrate **1310** into a plurality of substrates with each substrate **1310** having one weak spot **1312**. Alternatively, the substrates **1310** may be provided as individual pieces and one weak spot **1312** is formed onto one substrate **1310**. The weak spots **1312** may be formed on the substrates **1310** by printing the weak spots onto the substrates **1310**. The method **1100** further includes providing **904** a conductor separately from the plurality of substrates and the plurality of weak spots. The method **1100** also includes attaching **1106** the coplanar connector sections of the conductor to respective ones of the plurality of weak spots such that the plurality of weak spots may be longitudinally spaced apart from one another along the conductor and the plurality of planar substrates may be longitudinally spaced apart from one another along the conductor. In other words, the weak spots **1312** may be spaced apart from one another along a longitudinal direction of the conductor **314**, and the substrates **1310** may be spaced apart from one another along the longitudinal direction of the conductor **314**.

(63) In some embodiments, in attaching **1106** the weak spot **1312** to the conductor **314**, solder **1110**

is applied to the weak spot pads **1203** with the substrate **1310** initially as one piece. In one example, the solder **1110** is applied to the substrate **1310** by stencil printing the solder **1110** onto the substrate **1310** and reflowing the solder **1110** on the substrate **1310**. Afterwards, the weak spots **1312** may be separated from one another such that each substrate **1310** includes one weak spot **1312**. Weak spot pads **1203** may be placed in a tape and reel or matrix tray. A solder paste or flux (not illustrated) is applied to coplanar connector sections **318** of the conductor **314**. In one example, the solder paste or flux is placed on a side of the coplanar connector section **318** opposite a valley **1112** formed by the coplanar connector section **318** and its neighboring obliquely extending sections **320**. The conductor **314** is placed over the weak spot pads **1203**. Alternatively, the weak spot pads **1203** may be placed over the conductor **314**. In another example, the solder paste or flux is placed on a side of the coplanar connector section **318** the same as the valley **1112**. The weak spot pads **1203** may be picked up and placed in the valley **1112** of the conductor **314**. Once the weak spot pads **1203** and the conductor **314** are placed together, the conductor **314** and the weak spot pads **1203** may be reflowed. In one example, weight may be applied to the tops of the substrates **1310** or the coplanar connector section **318** to facilitate reflow.

(64) FIG. **12** is a schematic diagram of an exemplary embodiment of attaching **1106** the coplanar connector sections of the conductor, where the weak spots **1312** may be mounted to the side of the conductor **314** opposite the valley **1112**. The weak spot pads **1203** may be lined up with a spacing **1202** between neighboring weak spot pads **1203** such that a distance **1204** between neighboring weak spot pads **1203** is approximately the same as a distance **1206** between neighboring coplanar connector sections **318**.

(65) In the exemplary embodiment, the coplanar connector section **318** includes two portions **1209** separate by a gap **1210** (also see FIG. **10A**). The two portions **1209** may be connected by the support bridge **802**. The support bridge **802** extends from the two portions **1209** in the same direction as the extension direction of the obliquely extending section **320**. In mounting conductor **314** to the weak spot pads **1203**, the support bridge and the gap **1210** may be used to align the coplanar connector section **318** with the weak spot **1312** such that the gap **1210** is aligned with the openings **1208** (see FIG. **10A**). The support bridges **802** may be removed after the coplanar connector sections **318** of the conductor **314** are attached to the plurality of weak spots **1312**.

(66) FIGS. **13A-13C** show another embodiment of attaching **1106** the coplanar connector sections to the respective ones of the plurality of weak spots, where the plurality of weak spots **1312** may be mounted on the side of the coplanar connector sections **318** the same as the valley **1112**. FIG. **13A** is a top view of a fuse element assembly **1302-a**. FIGS. **13B** and **13C** are cross sectional views of the fuse element assembly **1302-a** along line **13B-13B** as marked in FIG. **13A**. FIG. **13C** shows an alternative configuration of the fuse element assembly **1302-a**. Different from the fuse element assembly **1302** (FIGS. **10A-10C**), where the weak spot pads **1203** may be disposed on a side of the coplanar connector sections **318** opposite the valleys **1112**, the weak spot pads **1203** in the fuse element assembly **1302-a** may be disposed in the valley **1112**. By disposing the weak spot pads **1203** in the valley **1112**, the arc is formed at the side opposite the weak spot **1312** or the valley **1112**, which is surrounded by arc extinguishing filler **330**, thereby increasing the effectiveness of arc quenching. The dimensions of the conductor **314** and/or the weak spot pads **1203** may be modified for the conductor **314** to receive the weak spot pads **1203** in the valleys **1112** of the conductor **314**. In some embodiments, the obliquely extending sections **320** and the support bridge **802** form a receptacle sized to receive the substrate **1310** therein. The oblique extending sections **320** and the support bridge **802** may be used to align and hold the substrates in position during reflow. The support bridges **802** may be removed after the coplanar connector sections **318** of the conductor **314** are attached to the plurality of weak spots **1312**.

(67) FIG. **13C** shows the alternative configuration of the fuse element assembly **1302-a**. Compared to the fuse element assembly **1302-a** illustrated in FIG. **13B**, the fuse element assembly **1302-a** illustrated in FIG. **13C** forms a pocket **1306** sized to receive the weak spot **1312** and weak spot

pads **1203**. As a result, the weak spots **1312** and the weak spot pads **1203** may be self-aligned in the coplanar connector sections **318**. In some embodiments, the support bridge **802** may be used for the alignment. Because of the self-alignment and/or increased ease in alignment, the connections between the weak spots **1312** and the coplanar connector sections **318** through soldering may be simplified or even eliminated.

(68) FIG. **14** is a schematic diagram of an exemplary coupling mechanism between the substrate **1310** and the coplanar connector section **318**. The coupling mechanism may be applied to the fuse element assembly **302**, **1302**, **1302-a**. In the exemplary embodiment, the weak spot **1312** is deposited on the substrate **1310**. A first solder **1402** and a second solder **1404** may be used to couple the substrate **1310** with the coplanar connector section **318**. The melting temperature of the first solder **1402** is higher than the melting temperature of the second solder **1404**. As a result, the first solder **1402** forms a diffusion barrier for the weak spot **312** to diffuse into the first and second solders **1402**, **1404**, and slows down melting of the first and second solders **1402**, **1404**, thereby reducing solder joint fatigue, which causes failure in electrical connections and renders fuses inoperable.

(69) Weak spots **312** may be used in place of weak spots **1312** and vice versa to enable the fuse element assemblies and methods to function or operate as described herein.

(70) The benefits and advantages of the present disclosure are now believed to have been amply illustrated in relation to the exemplary embodiments disclosed.

(71) Various embodiments of power fuses and fuse element assemblies and their fabrication methods are described herein including a plurality of weak spots formed on a substrate without stamped weak spot openings, thereby avoiding thermal-mechanical fatigue strain in the fuse element assembly when subjected to transient load current cycling events. Further, the fuse assembly includes a conductor having coplanar connector sections mounted on the weak spots and obliquely extending sections extending above the substrate such that an arc extinguishing filler can be disposed to surround at least part of the fuse element assembly, thereby effectively extinguishing arc generated after the fuse element assembly opens at predetermined current conditions.

(72) While exemplary embodiments of components, assemblies and systems are described, variations of the components, assemblies and systems are possible to achieve similar advantages and effects. Specifically, the shape and the geometry of the components and assemblies, and the relative locations of the components in the assembly, may be varied from those described and depicted without departing from inventive concepts described. Also, in certain embodiments, certain components in the assemblies described may be omitted to accommodate particular types of fuses or the needs of particular installations, while still providing the needed performance and functionality of the fuses.

(73) An embodiment of a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system has been disclosed. The power fuse includes at least one fuse element assembly that includes an elongated planar substrate, a plurality of fusible weak spots, and a conductor. The plurality of fusible weak spots are formed on the planar substrate and are longitudinally spaced from one another on the planar substrate. The conductor is separately provided from the planar substrate and the plurality of weak spots. The conductor includes a solid elongated strip of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to the transient load current cycling events. The solid elongated strip of metal includes coplanar connector sections that are mounted to respective ones of the plurality of weak spots on the planar substrate and obliquely extending sections bent out of plane of the connector sections to extend above the elongated planar substrate in between the plurality of fusible weak spots. The conductor further includes first and second terminal tabs that extend coplanar to one another in a plane parallel to but spaced from the connector sections and the substrate.

(74) Optionally, the power fuse further includes an arc quenching media that surrounds at least part

of the at least one fuse element assembly. The at least one fuse element assembly further includes a dielectric layer formed over the substrate and nested between the substrate and the plurality of weak spots. The conductor is formed in one piece. The substrate is alumina ceramic. The power fuse further includes a housing enclosing the at least one fuse element assembly. The plurality of fusible weak spots are printed on the planar substrate. The power fuse of has a voltage rating of at least 500 V. The power fuse has a current rating of at least 150 A. The at least one fuse element assembly includes first and second fuse element assemblies electrically connected in parallel with each other.

(75) A method of fabricating a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system has been disclosed. The method includes forming a plurality of fusible weak spots on an elongated planar substrate such that the plurality of fusible weak spots are longitudinally spaced from one another on the planar substrate. The method further includes providing a conductor separately from the planar substrate and the plurality of weak spots. The conductor includes a solid elongated strip of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to the transient load current cycling events. The solid elongated strip of metal includes coplanar connector sections and obliquely extending sections bent out of plane of the connector sections. The conductor further includes first and second terminal tabs that extend coplanar to one another. The method also includes mounting the coplanar connector sections of the conductor to respective ones of the plurality of weak spots on the planar substrate such that the obliquely extending sections of the conductor extend above the elongated planar substrate in between the plurality of fusible weak spots and the first and second terminal tabs extend coplanar to one another in a plane parallel to but spaced from the connector sections and the substrate, thereby completing a first fuse element assembly.

(76) Optionally, the method further includes surrounding at least part of the first fuse element assembly with an arc quenching medium. Forming a plurality of weak spots includes printing the plurality of weak spots on the elongated planar substrate. Forming a plurality of weak spots further includes providing a dielectric layer on the substrate, and forming the plurality of weak spots over the dielectric layer to cover the dielectric layer and to nest the dielectric layer between the substrate and the plurality of weak spots. Forming a dielectric layer includes printing the dielectric layer on the substrate, and forming the plurality of weak spots includes printing the plurality of weak spots over the dielectric layer to cover the dielectric layer and to nest the dielectric layer between the substrate and the plurality of weak spots. Providing a conductor further includes forming the conductor in one piece. The conductor is formed with support bridges connecting the coplanar connector sections, and mounting the coplanar connector sections further includes removing the support bridges after the coplanar connector sections of the conductor have been mounted on respective ones of the plurality of weak spots. The substrate includes alumina ceramic. The method further includes forming a second fuse element assembly, and electrically connecting the first and second fuse element assemblies in parallel with each other. The method further includes electrically connecting the first and second terminal tabs of the conductor with first and second conductive terminals, and enclosing the first fuse element assembly with a housing, leaving at least part of the first and second conductive terminals exposed.

(77) Another embodiment of a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system is disclosed. The power fuse includes at least one fuse element assembly including a plurality of planar substrates, a plurality of fusible weak spots, each formed on one of the plurality of planar substrates, and a conductor. The conductor is separately provided from the plurality of planar substrates and the plurality of weak spots, wherein the conductor includes an elongated strip of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to the transient load current cycling events. The elongated strip of metal further includes

coplanar connector sections that are attached to respective ones of the plurality of weak spots and obliquely extending sections bent out of plane of the coplanar connector sections. The plurality of weak spots are longitudinally spaced apart from one another along the conductor, and the plurality of planar substrates are longitudinally spaced apart from one another along the conductor.

(78) Optionally, one of the plurality of fusible weak spots includes openings. The plurality of fusible weak spots are printed on the planar substrates. Each of the plurality of weak spots is attached to a side of one of the coplanar connector sections the same as a valley formed by the coplanar connector section and its neighboring obliquely extending sections. The coplanar connector section forms a pocket sized to receive the weak spot therein. The coplanar connector sections are attached to respective ones of the plurality of weak spots through a first solder and a second solder, the first solder having a melting temperature higher than the melting temperature of the second solder, the first solder deposited over the respective ones of the plurality of weak spots, and the second solder deposited over the first solder.

(79) Another embodiment of a method of fabricating a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system is disclosed. The method includes forming a plurality of fusible weak spots on a plurality of planar substrates. The method also includes providing a conductor separately from the plurality of planar substrates and the plurality of weak spots, wherein the conductor includes an elongated strip of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to the transient load current cycling events. The elongated strip of metal includes coplanar connector sections and obliquely extending sections bent out of plane of the coplanar connector sections. The method further includes attaching the coplanar connector sections of the conductor to respective ones of the plurality of weak spots such that the plurality of weak spots are longitudinally spaced apart from one another along the conductor and the plurality of planar substrates are longitudinally spaced apart from one another along the conductor.

(80) Optionally, attaching the coplanar connector sections further includes attaching one of the plurality of weak spots to its respective one of the coplanar connector sections at a side of the coplanar connector section opposite a valley formed by the coplanar connector section and its neighboring obliquely extending sections. Alternatively, attaching the coplanar connector sections further includes attaching one of the plurality of weak spots to its respective one of the coplanar connector sections at a side of the coplanar connector section the same as a valley formed by the coplanar connector section and its neighboring obliquely extending sections. The conductor further includes a support bridge connecting the coplanar connector sections, the obliquely extending sections and the support bridge forming a receptacle sized to receive one of the plurality of planar substrates therein. Attaching the coplanar connector sections further includes aligning the coplanar connector sections with the plurality of planar substrates using the support bridges and the obliquely extending sections and holding the planar substrates in place using the support bridges and the obliquely extending sections during reflow. Attaching the coplanar connector sections also includes removing the support bridges after the coplanar connector sections of the conductor have been attached with respective ones of the plurality of weak spots. Forming a plurality of fusible weak spots further includes forming the plurality of fusible weak spots on a single piece of planar substrate and separating the single piece of planar substrate into the plurality of planar substrates such that each planar substrate includes one weak spot. Forming the plurality of fusible weak spots on a single piece of planar substrate further includes applying a first solder to the plurality of weak spots. Applying a first solder further includes stencil printing the first solder to the plurality of weak spots and reflowing the first solder on the plurality of weak spots. Attaching the coplanar connector sections further includes dispensing a second solder on the coplanar connector sections of the conductor, wherein the second solder has a melting temperature lower than the melting temperature of the first solder, placing the plurality of weak spots with the coplanar connector

sections such that the first solder and the second solder face each other, and reflowing the first solder and the second solder. Attaching the coplanar connector sections further includes placing the plurality of weak spots with the coplanar connector sections and applying weight to at least one of the plurality of planar substrates and the coplanar connector sections. One of the plurality of fusible weak spots includes openings. Forming a plurality of fusible weak spots further includes forming the plurality of fusible weak spots on the plurality of planar substrates by printing the plurality of fusible weak spots on the plurality of planar substrates. One of the coplanar connector sections forms a pocket sized to receive one of the plurality of weak spots, attaching the coplanar connector sections further including disposing the weak spot into the pocket.

(81) One more embodiment of a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system is disclosed. The power fuse includes at least one fuse element assembly including one or more substrates, one or more fusible weak spots each printed on one of the one or more substrates, and a conductor. The conductor is separately provided from the one or more substrates and the one or more weak spots, wherein the conductor includes an elongated strip of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to the transient load current cycling events. The elongated strip of metal further includes coplanar connector sections that are attached to respective ones of the one or more weak spots and obliquely extending sections bent out of plane of the coplanar connector sections. The one or more weak spots are longitudinally spaced apart from one another along the conductor, and the one or more substrates are longitudinally spaced apart from one another along the conductor.

(82) Optionally, one of the one or more substrates forms into a rod having an increased thickness than a substrate formed as a sheet.

(83) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

## Claims

1. A power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system, the power fuse comprising: at least one fuse element assembly comprising: one or more substrates; one or more sets of weak spots respectively formed on top of the one or more substrates, each set of weak spots comprising one or more fusible weak spots; and a conductor separately provided from the one or more substrates and the one or more sets of weak spots, wherein the conductor comprises one or more strips of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to transient load current cycling events, the conductor further comprising: connector sections that are attached to respective ones of the one or more sets of weak spots; and extending sections coupling the connector sections, wherein the one or more substrates are longitudinally spaced apart from one another along the conductor.
2. The power fuse of claim 1, wherein one of the one or more fusible weak spots includes openings.
3. The power fuse of claim 1, wherein one or more of the sets of weak spots are printed on respective ones of the one or more substrates.
4. The power fuse of claim 1, wherein each of the one or more sets of weak spots is attached to a side of one of the connector sections, wherein the one of the connector sections forms one or more

pockets sized to receive the set of weak spots therein.

5. The power fuse of claim 1, wherein the connector sections are attached to respective ones of the sets of weak spots through a first solder and a second solder, the first solder having a melting temperature higher than a melting temperature of the second solder, the first solder deposited over the respective ones of the sets of weak spots, and the second solder deposited over the first solder.

6. The power fuse of claim 1, wherein one of the one or more substrates forms into a rod having an increased thickness than a substrate formed as a sheet.

7. A method of fabricating a power fuse for protecting an electrical load subject to transient load current cycling events in a direct current electrical power system, the method comprising: forming one or more sets of weak spots respectively on top of one or more substrates, each set of weak spots comprising one or more fusible weak spots; providing a conductor separately from the one or more substrates and the one or more sets of weak spots, wherein the conductor includes one or more strips of metal having no stamped weak spot openings therein and therefore avoiding thermal-mechanical fatigue strain in the conductor when subjected to transient load current cycling events, the conductor further including: connector sections, and extending sections coupling the connector sections; and attaching the connector sections of the conductor to respective ones of the one or more sets of weak spots, wherein the one or more substrates are longitudinally spaced apart from one another along the conductor.

8. The method of claim 7, wherein attaching the connector sections further comprises attaching one of the sets of weak spots to a side of one of the connector sections, wherein the one of the connector sections forms one or more pockets sized to receive the one of the sets of weak spots therein.

9. The method of claim 7, wherein the conductor further includes one or more support bridges, each support bridge connecting pairs of connector sections, attaching the connector sections further comprising: aligning the connector sections with the one or more substrates using the one or more support bridges and the extending sections; holding the substrates in place using the support bridges and the extending sections during reflow; and removing the support bridges after the connector sections of the conductor have been attached with respective ones of the sets of weak spots.

10. The method of claim 7, wherein forming each of the one or more sets of weak spots further comprises: forming the one or more fusible weak spots on a single piece of respective ones of the one or more substrates; and separating the single piece of substrate into the one or more substrates such that each substrate includes one weak spot.

11. The method of claim 10, wherein forming the fusible weak spots on a single piece of substrate further comprises applying a first solder to the set of weak spots.

12. The method of claim 11, wherein applying a first solder further comprises: stencil printing the first solder to the set of weak spots; and reflowing the first solder on the set of weak spots.

13. The method of claim 11, wherein attaching the connector sections further comprises: dispensing a second solder on the connector sections of the conductor, wherein the second solder has a melting temperature lower than a melting temperature of the first solder; placing the set of weak spots with the connector sections such that the first solder and the second solder face each other; and reflowing the first solder and the second solder.

14. The method of claim 7, wherein attaching the connector sections further comprises: placing the set of weak spots with the connector sections; and applying weight to at least one of the one or more substrates and the connector sections.

15. The method of claim 7, wherein one of the one or more fusible weak spots includes openings.

16. The method of claim 7, wherein forming a set of weak spots further comprises forming the one or more fusible weak spots on the one or more substrates by printing the one or more fusible weak spots on the one or more substrates.

17. The method of claim 7, wherein one of the connector sections forms a pocket sized to receive

one of the sets of weak spots, attaching the connector sections further comprising disposing the one of the sets of weak spots into the pocket.

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